Abdulrahman Bahjat

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SUMMARY

Penn State mechanical engineering student with hands-on experience in CAD, robotics, 3D printing, and shipping full-stack web and mobile applications for real users.

EDUCATION

Penn State University · 2025 – present
B.S. Mechanical Engineering (in progress) · University Park, PA

- Engineering design (EDSGN 100), calculus, programming for engineers (CMPSC 200)

King Abdullah Academy · 2021 – 2025
High school diploma · IB Diploma Programme courses · Herndon, VA

- Robotics, digital society, and creative design electives

ACTIVITIES & PROJECTS

Web developer (CAS project) · 2024 – 2025
Sweet Moment Chocolates, Herndon, VA

- Designed and built the online storefront for a local handcrafted-chocolate business.
- Implemented catalog, cart, checkout, and an admin panel; later rebuilt as a fast static site.

Robotics program · 2022 – 2023
King Abdullah Academy

- Modeled three robot designs in CAD (Crab Bot, Scorpion, Spin Bot) with different chassis and mechanisms.

SELECTED PROJECTS

Sweet Moment Chocolates · 2024 – 2025
Online storefront for a local chocolatier
Technologies: React, TypeScript, Express, Stripe, Drizzle ORM, Tailwind CSS, Vite, GitHub Pages
Live: https://abdulrahmanb27.github.io/SweetMoment/ · Code: https://github.com/AbdulrahmanB27/SweetMoment

MyKitchen · 2026
Recipe dashboard as a PWA and native apps
Technologies: React 19, TypeScript, Tailwind CSS 4, Vite, Cloudflare Pages Functions, Cloudflare D1, Capacitor, Framer Motion, Gemini API
Live: https://abdulrahmanb27.github.io/MyKitchenCloudflare/ · Code: https://github.com/AbdulrahmanB27/MyKitchenCloudflare

VEX Robotics Designs · 2023
Three robot concepts modeled in CAD
Technologies: VEX Robotics, CAD, Mechanical design, 3D printing

Minecraft Adhan Clock · 2025
Raspberry Pi prayer clock styled after Minecraft
Technologies: Python, pygame, Raspberry Pi, Touchscreen UI, astral, hijri-converter

SKILLS

CAD & manufacturing: SolidWorks, Fusion 360, Autodesk Inventor, OpenSCAD, FDM 3D printing, Blender (basics)
Programming: Python, TypeScript, JavaScript, MATLAB, SQL
Web & mobile: React, Vite, Tailwind CSS, Node.js / Express, Cloudflare Pages & D1, Capacitor (iOS/Android), React Native / Expo, Stripe
3D & interactive: three.js, React Three Fiber, WebGL, GSAP
Hardware & robotics: VEX Robotics, Raspberry Pi, Arduino, Touchscreen UIs
Tools: Git / GitHub, VS Code, GitHub Pages, Replit, GIMP

Portfolio: https://abdulrahmanb27.github.io/AstroPortfolio/ · Updated 2026-09